



KEMENTERIAN PENDIDIKAN TINGGI, SAINS, DAN TEKNOLOGI
POLITEKNIK NEGERI MALANG
JURUSAN TEKNOLOGI INFORMASI
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Subject	:	Praktikum Algoritma
Major	:	D4 – D4 – Teknik Informatika
Semester	:	2
Class	:	II
NIM	:	254107020251
Name	:	Ringga Budi Utama

1. (Practicum 1: Class Declarations, Attributes and Methods)

```
import java.util.Scanner;

public class Mahasiswa {
    private String nim;
    private String nama;
    private String kelas;
    private double ipk;

    // Constructor
    public Mahasiswa(String nim, String nama, String kelas, double ipk) {
        this.nim = nim;
        this.nama = nama;
        this.kelas = kelas;
        this.ipk = ipk;
    }

    // Method to display data
    public void tampilkanData() {
        System.out.println("Data Mahasiswa:");
        System.out.println("NIM: " + nim);
        System.out.println("Nama: " + nama);
        System.out.println("Kelas: " + kelas);
        System.out.println("IPK: " + ipk);
    }

    // Method to calculate average
    public double tampilkanRataRata() {
        return ipk;
    }
}

// Test class
public class Main {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.println("Masukkan NIM:");
        String nim = input.nextLine();

        System.out.println("Masukkan Nama:");
        String nama = input.nextLine();

        System.out.println("Masukkan Kelas:");
        String kelas = input.nextLine();

        System.out.println("Masukkan IPK:");
        double ipk = input.nextDouble();

        Mahasiswa mahasiswa = new Mahasiswa(nim, nama, kelas, ipk);

        mahasiswa.tampilkanData();
        System.out.println("Rata-rata IPK: " + mahasiswa.tampilkanRataRata());
    }
}
```

1. Name two characteristics of a class or object!

Two characteristics of a class or object are:

- **Attributes** : variables that store data or information about the object.
- **Methods** : functions that define the behavior or actions of the object.

2. How many attributes does the Mahasiswa (Student) class have? Mention them.

The Mahasiswa class has **four attributes**, namely:

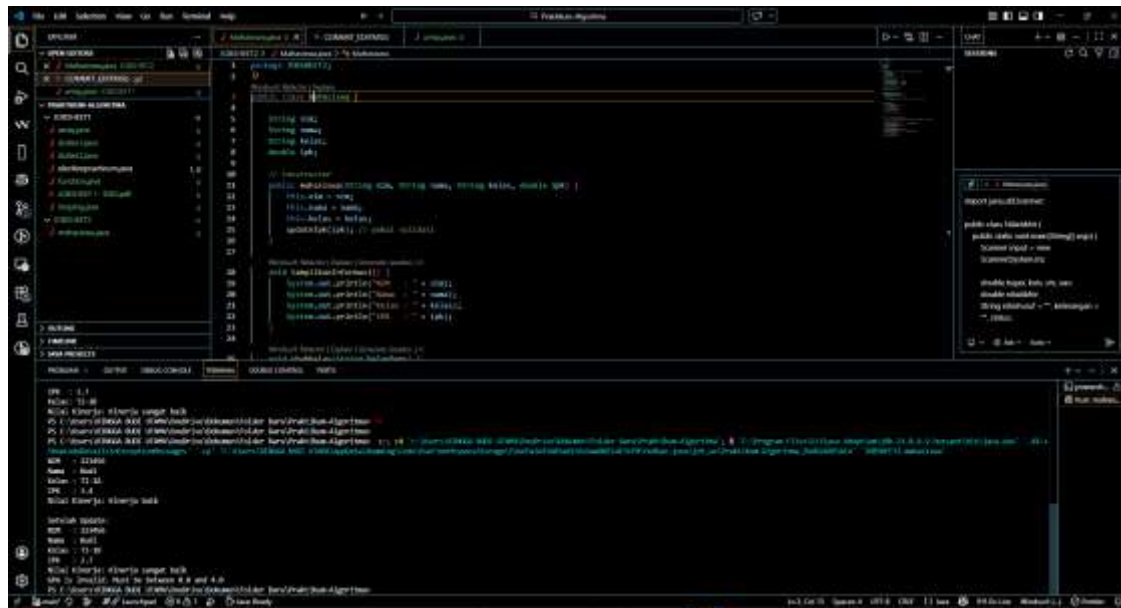
- nim
- nama
- kelas
- ipk

2. How many methods does the class have? Mention them.

The class has **four methods**, namely:

- tampilkanInformasi()
- ubahKelas(String kelasBaru)
- updateIpk(double ipkBaru)
- nilaiKinerja()

3.



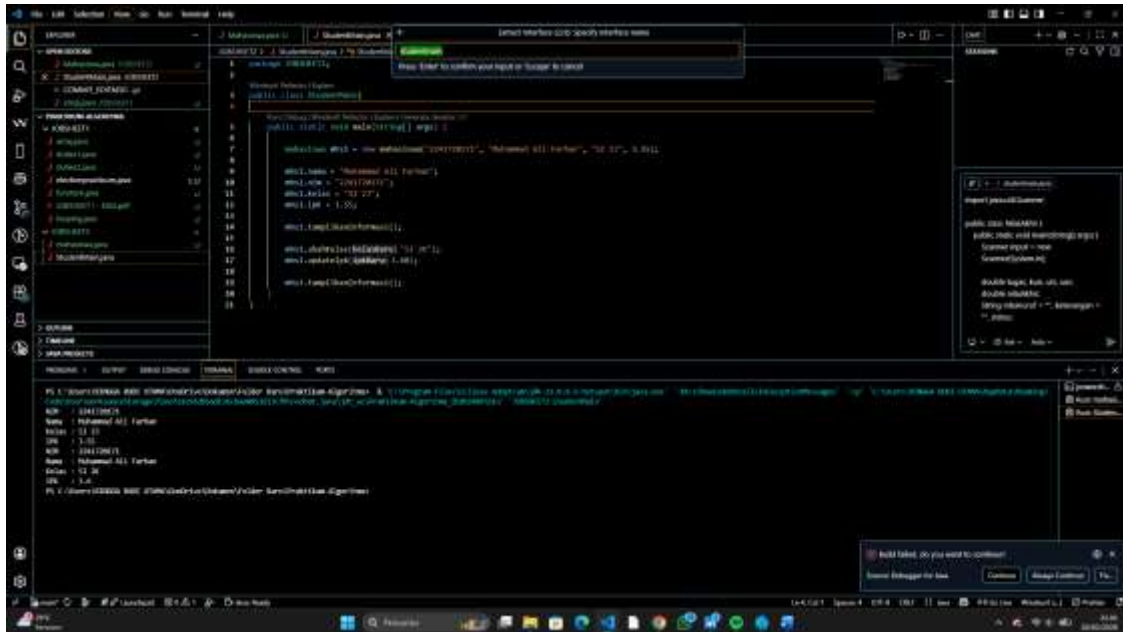
4. Explain how the nilaiKinerja() method works.

The `nilaiKinerja()` method evaluates student performance based on the GPA (ipk). It uses conditional statements (if-else) to determine the performance category.

The criteria are:

- $GPA \geq 3.5 \rightarrow$ "Very good performance"
- $GPA \geq 3.0 \rightarrow$ "Good performance"
- $GPA \geq 2.0 \rightarrow$ "Fair performance"
- $GPA < 2.0 \rightarrow$ "Poor performance"

2. Object Institutionalization, and Accessing Attributes and Methods



1. In the StudentMain class, indicate the line of program code used for the instance process! What is the name of the resulting object?

The line of code used for the instantiation process is:

- `Mahasiswa mhs1 = new Mahasiswa();`
- The name of the resulting object is **mhs1**.

2. How do I access the attributes and methods of an object?

Attributes and methods are accessed using the dot operator (.).

Example of accessing attributes:

- `mhs1.nama = "Muhammad Ali Farhan";`
- `mhs1.ipk = 3.55;`

Example of accessing methods:

- `mhs1.tampilkanInformasi();`
- `mhs1.updateIpk(3.60);`

3. Why are the output results of the method tampilInformasi() first and second different?

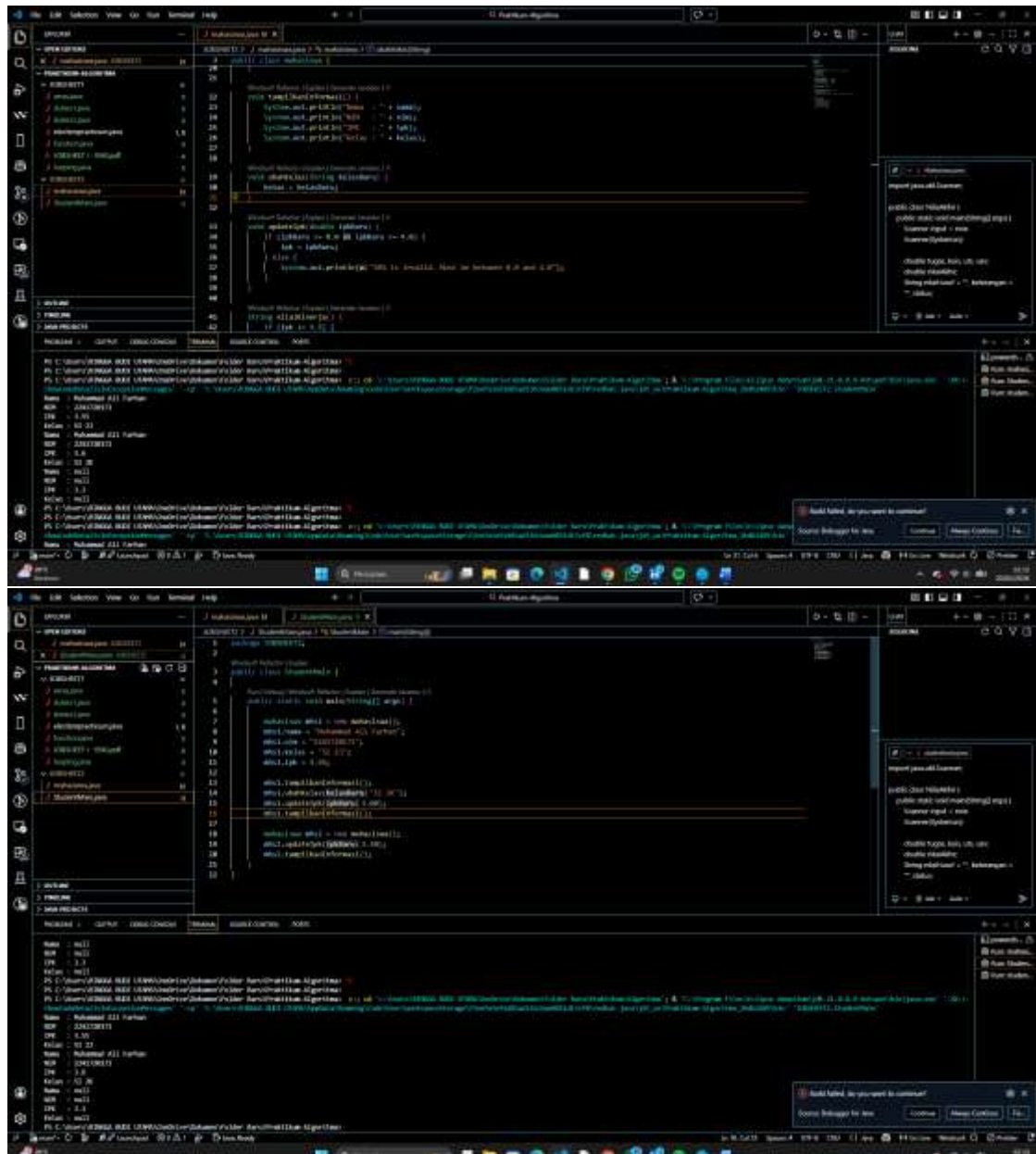
The outputs are different because the object's data was changed between the first and second method calls.

After the first tampilInformasi(), the program updates the class and GPA using:

- `mhs1.ubahKelas("SI 2K");`
- `mhs1.updateIpk(3.60);`

Therefore, the second tampilInformasi() displays the updated data.

3. Practicum 3: Constructor



1. In the Mahasiswa class in practicum 3, indicate the program code line used to declare a parameterized constructor!

The line of code used to declare the parameterized constructor is:

public Mahasiswa(String nama, String nim, double ipk, String kelas)

2. In the StudentMain class, what exactly does the following line of program do?

Mahasiswa mhs2 = new Mahasiswa("Annisa Nabila", "2141720160", 3.25, "TI 2L");

This line creates a new object named **mhs2** using the parameterized constructor.

It directly initializes the attributes nama, nim, ipk, and kelas with the given values at the time the object is created.

3. Remove the default constructor from the Mahasiswa class, then compile and run the program. How did it turn out? Explain why this is the case!

If the default constructor is removed and the program still contains:

```
Mahasiswa mhs1 = new Mahasiswa();
```

the program will produce a compilation error.

This happens because there is no constructor without parameters available anymore.

Since only the parameterized constructor exists, every object must be created using arguments.

4. After instantiating the object, do the methods in the Mahasiswa class have to be accessed sequentially? Explain why!

No, the methods do not have to be accessed sequentially.

Methods can be called in any order depending on the program's needs, because each method works independently unless it depends on changes made by another method.

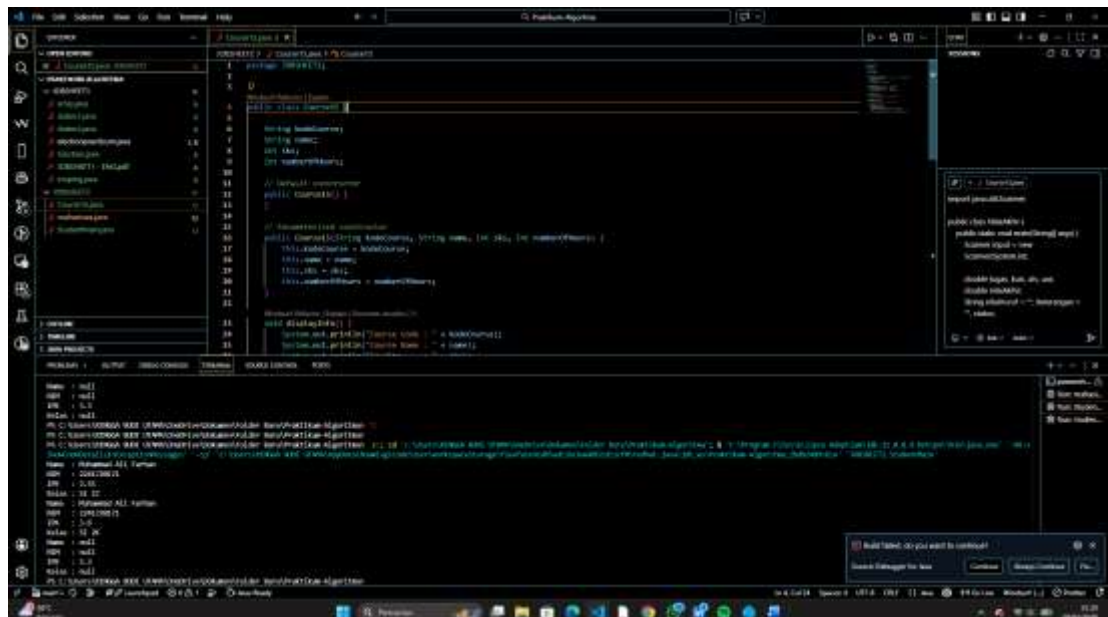
5. Create a new object with the name mhs<StudentName> using the parameterized constructor!

Example:

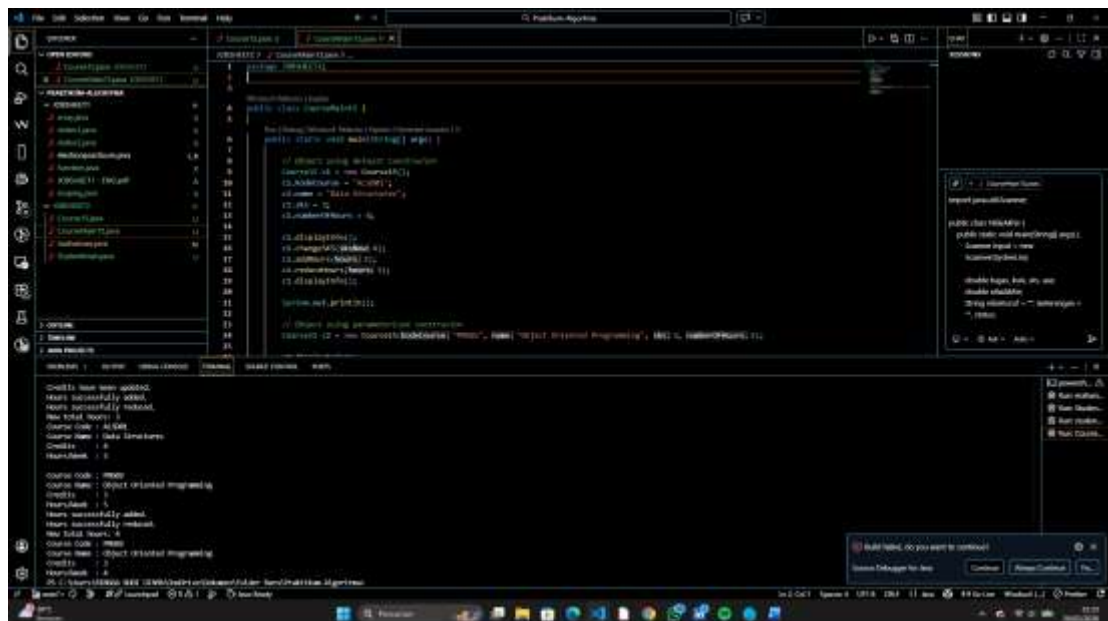
```
Mahasiswa mhsRingga = new Mahasiswa("Ringga Budi  
Utama", "2241720001", 3.50, "TI 2A");
```

This creates a new object using the parameterized constructor and directly assigns the attribute values.

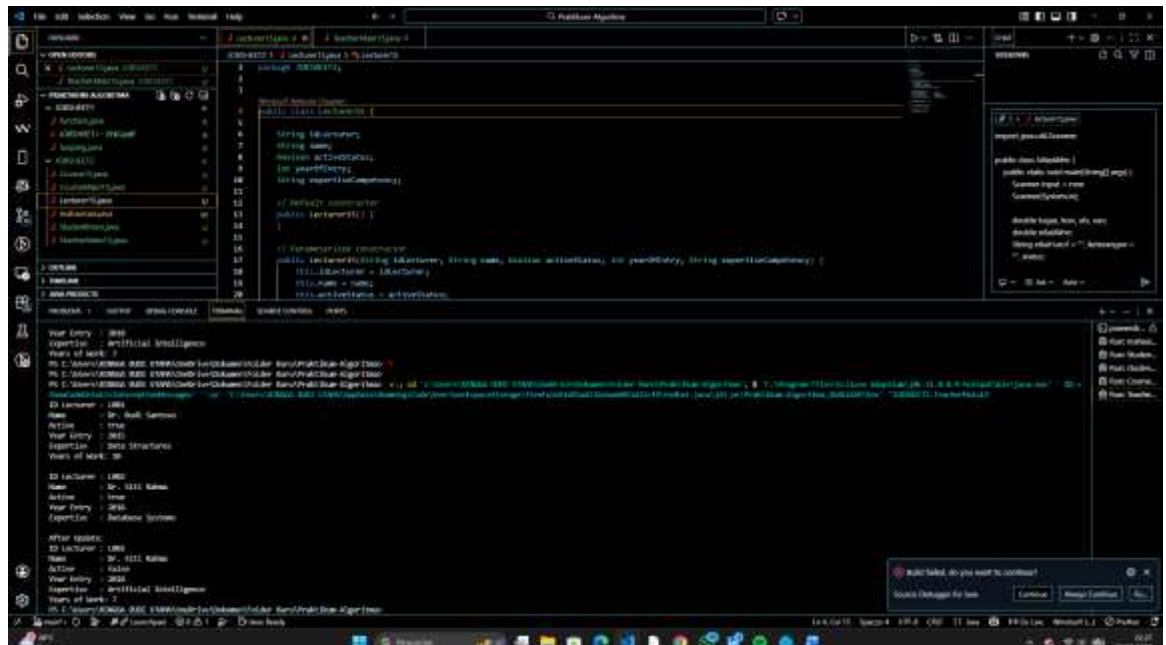
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